

VLAN Implementation Lab

LAB REPORT: VLAN Implementation

Tool Download Link:

<https://www.netacad.com/articles/news/download-cisco-packet-tracer>

Experiment No. 3

Title

Implementation of Virtual Local Area Networks (VLANs) and Inter-VLAN Communication

Objective

To configure VLANs on a switch, assign ports to different VLANs, and enable communication between VLANs using Router-on-a-Stick configuration.

Theory

A Virtual Local Area Network (VLAN) is a logical grouping of devices within a network regardless of their physical location. VLANs improve network performance, enhance security, and simplify network management by separating broadcast domains.

Benefits of VLANs:

- Improved security
- Reduced broadcast traffic
- Better network management
- Logical segmentation of users and departments

Inter-VLAN communication requires a Layer 3 device such as a router or Layer 3 switch.

Network Topology

VLAN 10 (SALES)

- PC1 – 192.168.10.11
- PC2 – 192.168.10.12

VLAN 20 (HR)

- PC3 – 192.168.20.11
- PC4 – 192.168.20.12

Gateway Addresses

- VLAN 10 Gateway: 192.168.10.1
- VLAN 20 Gateway: 192.168.20.1

Devices Used:

- 1 Cisco Switch
 - 1 Cisco Router
 - 4 PCs
 - Ethernet Cables
-

Procedure

Step 1: Create VLANs

```
enable
configure terminal
```

```
vlan 10
name SALES
```

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 10
Switch(config-vlan)#name SALES
Switch(config-vlan)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console
Switch#
```

```
vlan 20
name HR
```

```
end
```

```
Switch(config)#vlan 20
Switch(config-vlan)#name HR
Switch(config-vlan)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console
```

Step 2: Assign Ports to VLANs

VLAN 10

```
interface range fa0/1 - 2
switchport mode access
switchport access vlan 10
```

```
Switch#
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface range fa0/1-2
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 10
Switch(config-if-range)#no shutdown
Switch(config-if-range)#
```

VLAN 20

```
interface range fa0/3 - 4
switchport mode access
switchport access vlan 20
```

```
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#interface range fa0/3-4
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 20
Switch(config-if-range)#no shutdown
Switch(config-if-range)#
```

Step 3: Configure Trunk Port

```
interface fa0/24
switchport mode trunk
```

```
Switch(config)#interface fa0/24
Switch(config-if)#switchport mode trunk
Switch(config-if)#
```

Step 4: Configure Router-on-a-Stick

```
enable
configure terminal
```

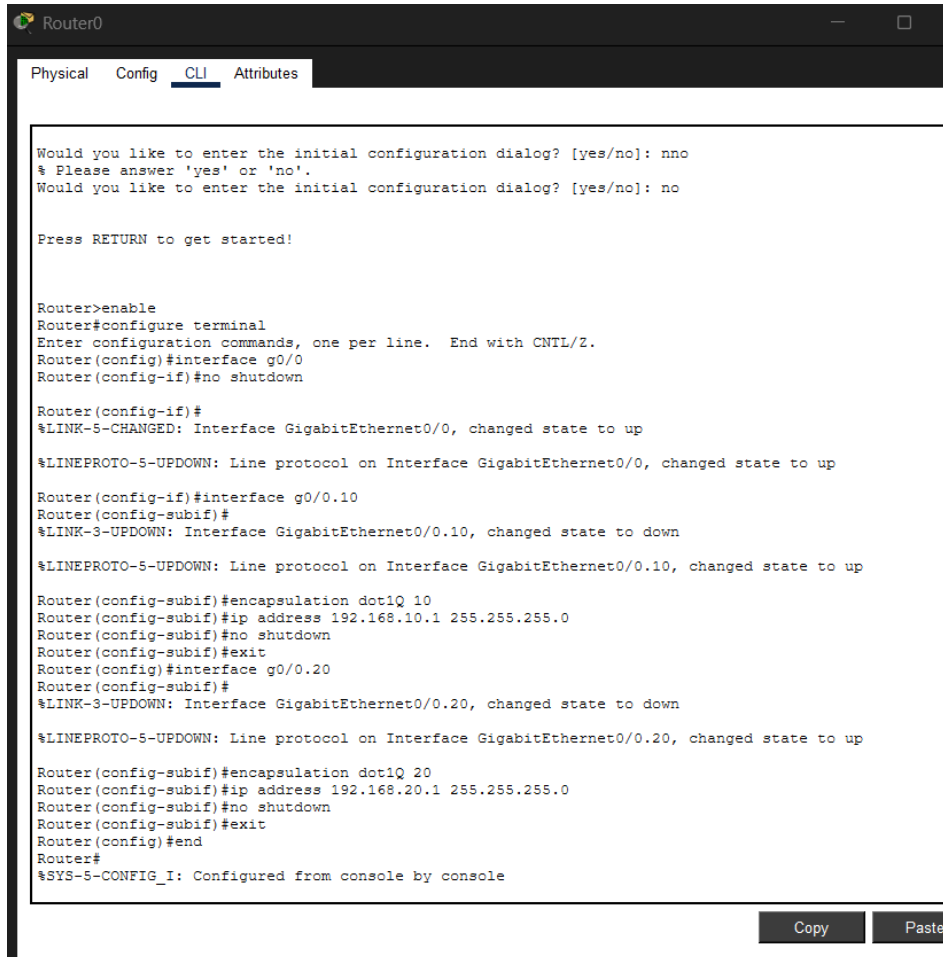
```
interface g0/0
no shutdown
```

```
interface g0/0.10
encapsulation dot1Q 10
```

```
ip address 192.168.10.1 255.255.255.0
```

```
interface g0/0.20  
encapsulation dot1q 20  
ip address 192.168.20.1 255.255.255.0
```

```
end
```



The screenshot shows a Cisco Router CLI window titled "Router0". The "CLI" tab is selected. The terminal output shows the following commands and responses:

```
Would you like to enter the initial configuration dialog? [yes/no]: nno  
% Please answer 'yes' or 'no'.  
Would you like to enter the initial configuration dialog? [yes/no]: no  
  
Press RETURN to get started!  
  
Router>enable  
Router#configure terminal  
Enter configuration commands, one per line. End with CNTL/Z.  
Router(config)#interface g0/0  
Router(config-if)#no shutdown  
  
Router(config-if)#  
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up  
  
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up  
  
Router(config-if)#interface g0/0.10  
Router(config-subif)#  
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.10, changed state to down  
  
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.10, changed state to up  
  
Router(config-subif)#encapsulation dot1q 10  
Router(config-subif)#ip address 192.168.10.1 255.255.255.0  
Router(config-subif)#no shutdown  
Router(config-subif)#exit  
Router(config)#interface g0/0.20  
Router(config-subif)#  
%LINK-3-UPDOWN: Interface GigabitEthernet0/0.20, changed state to down  
  
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0.20, changed state to up  
  
Router(config-subif)#encapsulation dot1q 20  
Router(config-subif)#ip address 192.168.20.1 255.255.255.0  
Router(config-subif)#no shutdown  
Router(config-subif)#exit  
Router(config)#end  
Router#  
%SYS-5-CONFIG_I: Configured from console by console
```

At the bottom right of the terminal window, there are "Copy" and "Paste" buttons.

Step 5: Configure PC IP Addresses

VLAN 10

PC IP Address

PC1 192.168.10.11

PC2 192.168.10.12

VLAN 20

PC IP Address

PC3 192.168.20.11

PC IP Address
PC4 192.168.20.12

PING

PC1 To PC2

```
^C
C:\>ping 192.168.10.12

Pinging 192.168.10.12 with 32 bytes of data:

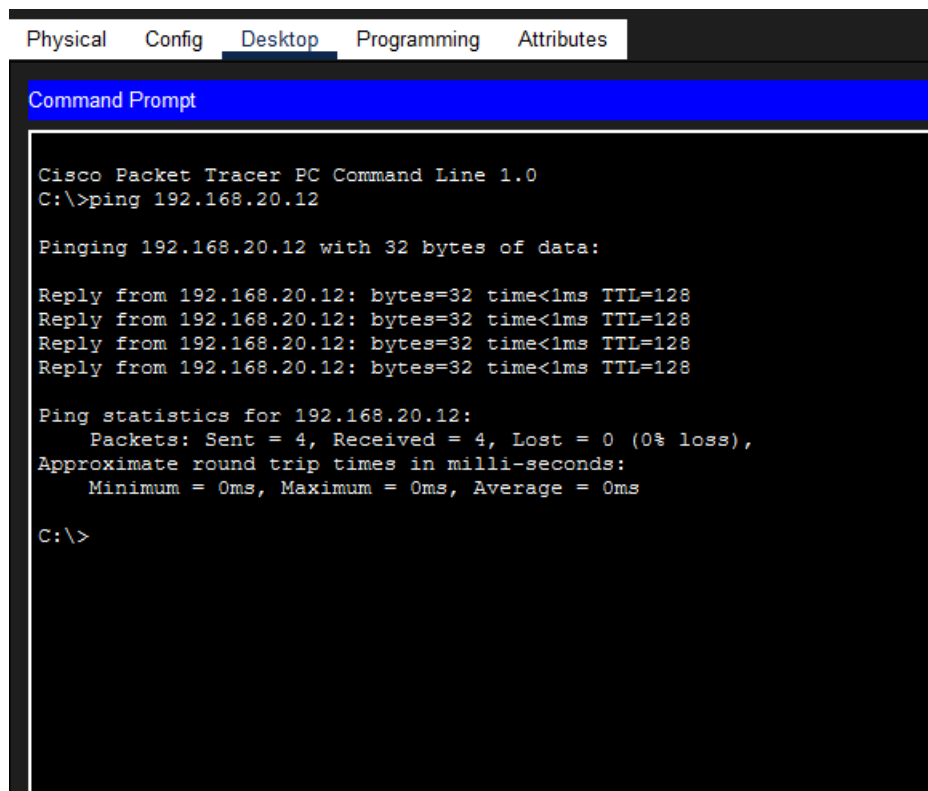
Reply from 192.168.10.12: bytes=32 time<1ms TTL=128
Reply from 192.168.10.12: bytes=32 time<1ms TTL=128
Reply from 192.168.10.12: bytes=32 time<1ms TTL=128
Reply from 192.168.10.12: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.12:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

NOW

PC3 To PC4



Physical Config **Desktop** Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.12

Pinging 192.168.20.12 with 32 bytes of data:

Reply from 192.168.20.12: bytes=32 time<1ms TTL=128
Reply from 192.168.20.12: bytes=32 time<1ms TTL=128
Reply from 192.168.20.12: bytes=32 time<1ms TTL=128
Reply from 192.168.20.12: bytes=32 time<1ms TTL=128

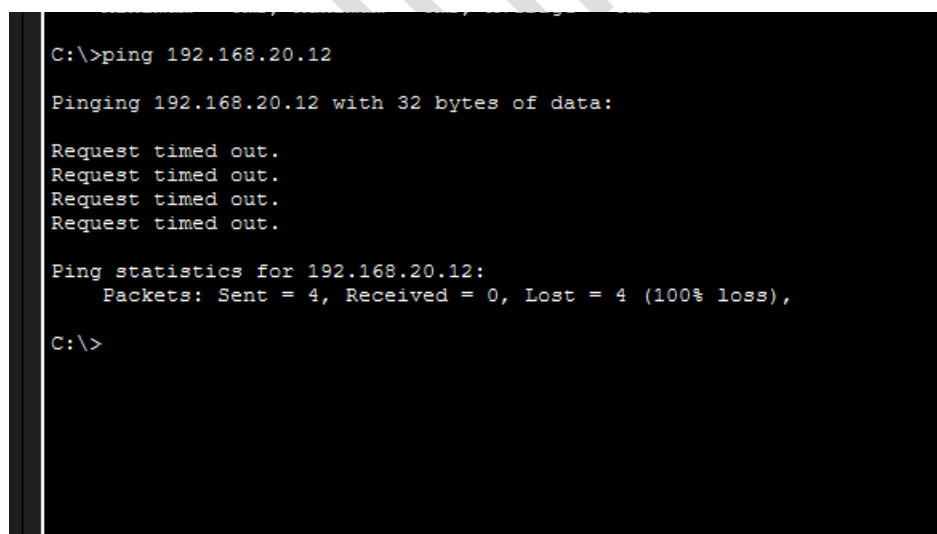
Ping statistics for 192.168.20.12:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Vlan Work Successfully.

Now

Ping PC1 To PC4



```
C:\>ping 192.168.20.12

Pinging 192.168.20.12 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.20.12:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

PC1 not Ping to PC4

Sales Department cannot ping to HR Department.

Before Inter-VLAN Routing

- VLAN 10 and VLAN 20 are separate broadcast domains.
- Devices in the same VLAN can communicate.
- Devices in different VLANs cannot communicate.

Example:

- PC1 (192.168.10.11) → PC2 (192.168.10.12)  Success
- PC1 (192.168.10.11) → PC4 (192.168.20.11)  Failed

```
C:\>ping 192.168.20.12

Pinging 192.168.20.12 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.20.12:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\>
```

After Inter-VLAN Routing

Using a router (Router-on-a-Stick) or a Layer 3 switch:

- PC1 (VLAN 10) → PC4 (VLAN 20)  Success
- PC2 (VLAN 10) → PC5 (VLAN 20)  Success

```
C:\>ping 192.168.20.12

Pinging 192.168.20.12 with 32 bytes of data:

Request timed out.
Reply from 192.168.20.12: bytes=32 time=1ms TTL=127
Reply from 192.168.20.12: bytes=32 time=11ms TTL=127
Reply from 192.168.20.12: bytes=32 time=10ms TTL=127
```

Verification Commands

```
show vlan brief
show interfaces trunk
```

```
show running-config  
show ip interface brief
```

Observations

1. VLAN 10 and VLAN 20 were successfully created.
 2. Ports were assigned correctly to their respective VLANs.
 3. Devices within the same VLAN communicated successfully.
 4. Devices in different VLANs could not communicate before router configuration.
-

Results

The VLAN implementation was completed successfully. VLAN 10 and VLAN 20 were created and configured on the switch. Inter-VLAN routing was established using a router, allowing all four PCs to communicate across VLAN boundaries.

Conclusion

This lab demonstrated the creation and management of VLANs in a switched network. VLANs effectively separated network traffic into different broadcast domains. Router-on-a-Stick configuration enabled communication between VLANs, illustrating the importance of Layer 3 devices in segmented networks.